Quick Bootstrapping of a Personalized Gaze Model from Real-Use Interactions

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Understanding human visual attention is essential for understanding human cognition, which in turn benefits human–computer interaction. Recent work has demonstrated a Personalized, Auto-Calibrating Eye-tracking (PACE) system, which makes it possible to achieve accurate gaze estimation using only an off-the-shelf webcam by identifying and collecting data implicitly from user interaction events. However, this method is constrained by the need for large amounts of well-annotated data. We thus present fast-PACE, an adaptation to PACE that exploits knowledge from existing data from different users to accelerate the learning speed of the personalized model. The result is an adaptive, data-driven approach that continuously "learns" its user and recalibrates, adapts, and improves with additional usage by a user. Experimental evaluations of fast-PACE demonstrate its competitive accuracy in iris localization, validity of alignment identification between gaze and interactions, and effectiveness of gaze transfer. In general, fast-PACE achieves an initial visual error of 3.98 degrees and then steadily improves to 2.52 degrees given incremental interaction-informed data. Our performance is comparable to state-of-the-art, but without the need for explicit training or calibration. Our technique addresses the data quality and quantity problems. It therefore has the potential to enable comprehensive gaze-aware applications in the wild.

CCS Concepts: • **Human-centered computing → Human computer interaction (HCI)**; • **Computing methodologies → Artificial intelligence**; **Computer vision**; *Computer vision tasks*;

Additional Key Words and Phrases: Gaze estimation, implicit modeling, data validation, gaze-interaction alignment, gaze transfer learning

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1 INTRODUCTION

Gaze information reflects human attention and cognition. It has great potential applications in numerous domains, such as education, crowdsourcing, and diagnostics. However, with few

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exceptions, most gaze estimation methods require calibration and nonperiodical recalibration to accommodate lighting and head pose variations, which is obviously cumbersome and inconvenient in real use [34]. For infrared eye trackers, this is just an inconvenience and usability issue, but it is much more critical for webcam-based gaze estimation systems. This is because a webcam-based system is more susceptible to changes in environment and appearance and therefore requires a large amount of user-specific calibration data to achieve reasonable accuracy for interaction [9]. Clearly, it is infeasible to demand lengthy daily calibration from users in a real-world system.

In the past few years, much research has investigated user behavior in human–computer interaction [46, 47, 48]. Since there is a strong correlation between eye gaze and interactional cues, such as cursor and caret locations, it should be possible to establish a mapping between gaze features and gaze point. We aim to achieve implicit gaze learning from collecting daily computer interactions with conventional off-the-shelf equipment.

A number of studies have demonstrated a correlation between gaze and cursor [11, 21], but there is little work that uses noisy daily interaction data for webcam gaze learning. Most previous works [6, 18, 26, 33] assume that users are looking where they click, or, to put it more broadly, that users are looking at the interaction cue at the moment that the interaction is triggered. However, this assumption may not be valid in real-use situations due to factors such as mind-absence, response delay, individuality, and task difference. It is therefore important to identify and use only reliable training instances for gaze learning, which can only be achieved by correctly understanding the temporal and spatial alignments between gaze and interactional cues.

Although the identification of alignments improves the data *quality* for the implicit gaze learning, data *quantity* also plays an essential role for a well-performing gaze model [39]. Collecting sufficient data through interactions can be unobtrusive, but it tends to be a slow process, and a good amount of time may be required for satisfactory performance. Inspired by the work on pursuit-based gaze interaction [38] that uses the motion correlation between gaze and visual target, we address the data quantity problem through an efficient gaze transfer learning technique that utilizes the gaze correlation among models of different individuals.

This article presents a fully developed technique that extends the previous state-of-the-art Personalized, Auto-Calibrating Eye-tracking (PACE) system [13] with novel contributions to build *fast*-PACE. Specifically, PACE focuses on the identification and validation of appropriate interaction-generated data points for training, while fast-PACE introduces (i) an improved method to locate the iris center accurately and (ii) a transfer learning method which efficiently utilizes limited interaction-informed data to rapidly construct a well-performing personalized gaze tracking model. We also conduct a rigorous evaluation on a larger and more realistic dataset to study the relevant issues of learning from real-use interactional cues.

The overall contributions of this article are as follows. We (i) propose a robust method for iris center localization to enrich PACE modeling; (ii) present an adaptation to PACE that relaxes its relatively restrictive and rigid constraints and creates an efficient, unobtrusive, adaptive, interaction-informed method that identifies gaze-interaction alignment in daily computer use; (iii) suggest an efficient strategy for gaze transfer learning given limited target data; and (iv) conduct an in-depth experimental study to investigate the effectiveness of fast-PACE from several aspects.

The remaining article is organized as follows: Section 2 provides a summary of related research work. Our system overview is described in Section 3. Sections 4 and 5 present our approaches for iris localization, interaction data validation, and gaze transfer. Experiments and results are covered in Section 6. Finally, we conclude this article and discuss the future work in Section 7.

2 RELATED WORK

This section begins with a discussion of the background of webcam-based gaze learning. We then describe recent research efforts on implicit data collection for gaze learning, followed by work on generalizing from limited data.

2.1 Webcam-Based Gaze Learning

A good number of studies have investigated robust techniques for eye geometry extraction from webcam signals to address environmental noise issues. Zhu and Yang [55] used the vector from the inner eye corner to the fitted iris center as an indicative feature. Cheung and Peng [3] added head pose information to the same vector to accommodate pose variations. In a similar manner, Valenti et al. [36] proposed a pose-retargeted gaze estimation method based on the displacement of the eye center with respect to head pose change. Wood and Bulling [41] calculated the optical axis based on the fitted ellipse of the iris limbus and estimated gaze by trigonometric approximation. However, simple rule-based eye feature extraction in the wild can be unreliable.

Data-driven approaches have been explored for eye geometric feature extraction. Huang et al. [14] suggested an eye alignment model for the localization of a set of eye and iris landmarks. However, training a reliable nonrigid alignment model needs numerous landmark annotations. Wood et al. [40] leveraged computer graphics techniques to synthesize various eye appearances, and constructed the alignment model based on the generated data. Despite prior successes in robust eye feature extraction, the acquisition of sufficient user-specific data for gaze estimation is still a challenging research problem.

There have been efforts made in using a large amount of data for gaze learning. Huang et al. [17] prepared a large dataset called TabletGaze, which contains more than 100,000 images. Zhang et al. [51] trained a Convolution Neural Network (CNN) on the MPIIGaze dataset, which contains more than 200,000 images, to recognize the gaze angle vector. Xu et al. [43] presented the TurkerGaze crowdsourcing platform, which was used to collect a large-scale gaze dataset. Krafka et al. [20] also used a CNN to model gaze from 1.5 million frames of face and eye regions. Similarly, Zhang et al. [53] presented a CNN that predicts gaze based only on full-face images. To speed up the data acquisition process, Wood et al. [39] rendered a million realistic eye images for gaze angle estimation. However, computer-generated data may not fully coincide with real user-specific data.

2.2 Implicit Collection of Gaze Data

To mitigate the annotation effort and obtain real user data, researchers have proposed to collect personal data implicitly through interactions. Sugano et al. [32] recorded users watching a video and estimated the gaze coordinates based on the saliency map of the video frames; however, how well the image saliency matches with the real gaze pattern is influenced by the complexity and semantics of visual stimuli. Hornof and Halverson [10] suggested a strategy that looks for interactions with known fixation points to use for recalibrating the eye gaze model, which relies on the knowledge of the visual context, such as target locations and layout irregularity.

Some studies exploited interaction for gaze estimation and learning. Fares et al. [6] used click data to correct gaze tracking results. Sugano et al. [33] used the mouse-click locations as ground truth data to update the gaze model incrementally. Similarly, Papoutsaki et al. [26] presented a browser-based eye tracker that learns from mouse-clicks and mouse movements. However, they rely on an assumption that users are gazing where they click or type, which has been shown to not always hold in real use [14]. Liebling et al. [21] showed that gaze and mouse coordination carries complex and subtle characteristics in real-use scenarios. Huang et al. [12] pointed out that there is a certain correlation between gaze and cursor, but with substantial variation.

To ensure the consistency between gazes and interactions, Sugano et al. [34] trained a gaze model on the eye images with modified click location, which is a Gaussian-weighted average of the locations of previous neighboring samples. Huang et al. [13] used the temporal and spatial alignments between keypresses, mouse-clicks from interaction logs, and gazes from the gaze signals to collect reliable training data to build a user-specific model. However, these approaches are time-consuming, and it takes a while to collect enough training data before good performance can be achieved. Therefore, we propose a transfer learning technique that addresses this time-consuming user data collection problem and accelerates the model adaptation process.

2.3 Generalizing from Limited Data

Making good use of limited personal data is the key to improving performance while reducing the calibration effort. Lu et al. [22] introduced a decomposition scheme to correct the head pose biases for gaze estimation. Their calibration requires users to rotate their heads while fixating on each calibration point. Lu et al. [23] later proposed to synthesize training samples for unseen head poses from images under certain reference head positions. Although these methods reduce the amount of calibration data needed, they require a specialized and explicit calibration procedure.

Transfer learning has recently become popular for knowledge adaptation and has been used to address the target data scarcity issue in different problems [25]. Many transfer learning studies focus on scenarios in which annotations of the target user are unavailable. In such cases, knowledge transfer mainly relies on instance reweighting, which can be computationally expensive and may also be unreliable.

There has been some works done in exploiting transfer with limited target data. Sangineto et al. [30] presented a method to alleviate the computational cost of facial affect transfer by using a regression function to identify the parameters of the target model according to the mapping from data distribution of source subjects into the parameters of their classifiers. Zen et al. [49] further simplified the mapping by using parameter transfer based on the support vectors. These personalization strategies produce promising results without expensive computation and therefore have application potential in the wild. However, parameter transfer requires that all involved classifiers share the same form of mapping function. This constrains their modeling capability. In contrast, Sugano et al. [35] proposed to learn an additional regression function that corrects the estimation of the source model. In a similar manner, Yao and Doretto [45] suggested the adaptation from multiple sources by reweighting the source classifiers according to their performances on the target data. Chen et al. [2] demonstrated the effectiveness of model reweighting in personalized facial affect modeling. Unlike prior studies, we adopted the correlation coefficient between the ground truth and the estimation of multiple source classifiers as the metric for model reweighting, which is a fast approach that is computationally lightweight and suitable for gaze transfer in practice.

3 SYSTEM OVERVIEW OF FAST-PACE

Figure 1 gives an overview of our approach. The green dashed frame indicates the PACE system [13], and the blue frame shows the novel fast-PACE framework. The goal of fast-PACE is to learn a practical, user-adaptive gaze model that uses a novel transfer learning approach to auto-update for the target user in an implicit and continuous manner and at a rapid pace. In addition, fast-PACE also enhances the PACE algorithms for better performance, including a more accurate and robust iris localization algorithm.

In order for fast-PACE to be feasibly integrated into real-use computing systems, we unobtrusively identify and collect training instances from daily interaction data. Our investigation focuses on four commonly encountered interaction activities, as shown in Table 1.

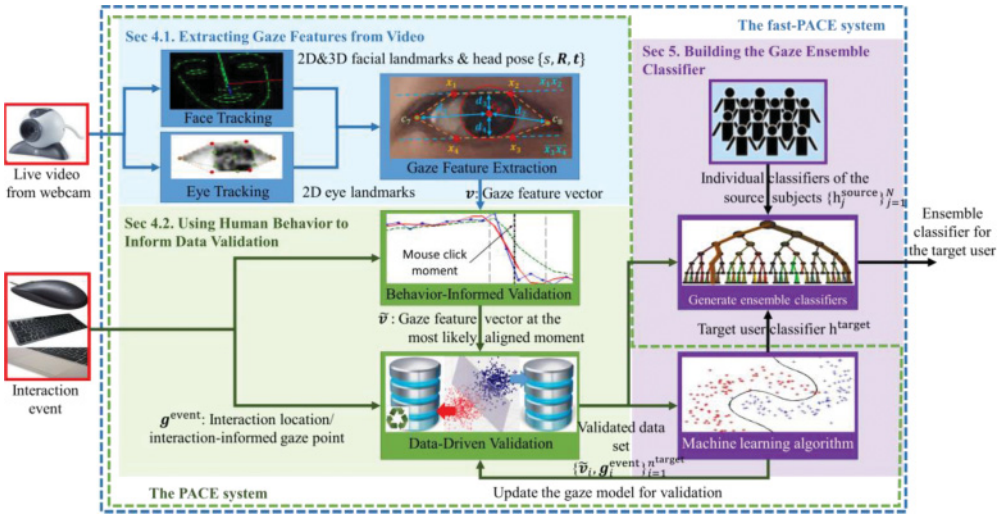


Fig. 1. System overview of fast-PACE. The green dashed frame indicates the components in PACE that learn gaze from aligned interaction data. The other two components inside the blue dashed frame denote the gaze transfer mechanism that adapts the source knowledge to gaze learning of the target user. The gaze transfer ensures that fast-PACE produces an acceptable gaze estimation with only limited target data.

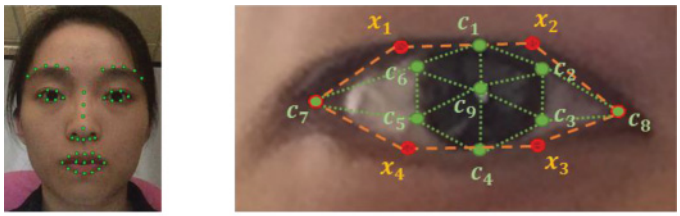
Table 1. Examples of Gaze Patterns from Common Interaction Behaviors

Interaction Event	Potential Gaze Pattern	Human Intention
Mouse click	Fixation on the mouse cursor	Link or button selection
Mouse double-click	Fixation on the mouse cursor	Word selection in document editing
Mouse button up after drag	Fixation on the mouse cursor	Paragraph selection in document editing
Keyboard letter key down	Fixation/smooth pursuit on the typing caret	Word typing

Section 4.1 (area shaded blue in Figure 1) discusses the gaze feature extraction from video. A standard webcam captures video of the user while mouse movements and keystrokes are tracked by the system. The gaze feature vector $\boldsymbol{v}$ is extracted based on the landmarks identified by the face and eye alignment models. If there are no incoming interaction events, the system estimates gaze points based on the existing gaze model and gaze feature vector $\boldsymbol{v}$.

Section 4.2 (area shaded in green) discusses the gaze data validation mechanism based on human behaviors. Upon the trigger of an interaction event, gaze feature vectors from a 3-second window preceding the interaction and the interaction-informed gaze point $\boldsymbol{g}^{\text{event}}$ are sent to a *behavior-informed validation* engine and a *data-driven validation* engine. The behavior-informed validation engine selects one vector, $\tilde{\boldsymbol{v}}$, that corresponds to the moment when the user's gaze is most likely to be aligned with the interaction event. The data-driven validation engine then further checks the validity of $\tilde{\boldsymbol{v}}$ based on the previous training samples. If $\tilde{\boldsymbol{v}}$ passes both of these validation steps, $[\tilde{\boldsymbol{v}}, \boldsymbol{g}^{\text{event}}]$ will be used as *target training data* to update the gaze regression model h^{target}. Otherwise, the data are retained for re-evaluation after the next update.

Section 5 (area shaded in purple) presents our gaze ensemble classifier. To shorten the learning period of the gaze model and thus make it more feasible for real-use situations, the validated



(a) (b)

Fig. 2. The landmarks tracked by the face and eye alignment models; (a) shows the 48 face landmarks, which are tracked by the SDM face alignment model, (b) shows an eye image and the eye landmarks $\boldsymbol{c}_1, \ldots, \boldsymbol{c}_9$. The dashed polygon indicates the eye contour, which is defined by the related landmarks from the face alignment model ($\boldsymbol{x}_1, \ldots, \boldsymbol{x}_4$) and the eye alignment model ($\boldsymbol{c}_7, \boldsymbol{c}_8$).

target training data, $\{\tilde{\boldsymbol{v}}_i, \boldsymbol{g}_i^{\text{event}}\}_{i=1}^{n^{\text{target}}}$, is used to learn the target classifier from the existing source classifiers $\{\text{h}_j^{\text{source}}\}_{j=1}^{N}$, which are prepared in advance from a set of source subjects. Here, n^{target} is the number of validated target samples, and N is the number of source subjects. The final gaze model of the user is the ensemble of the source classifiers $\{\text{h}_j^{\text{source}}\}_{j=1}^{N}$ and the target classifier h^{target}.

4 EXTRACTING ALIGNED FEATURES FOR GAZE ESTIMATION

This section describes the gaze feature extraction and the gaze-interaction alignment in fast-PACE.

4.1 Extracting Gaze Features from Video

4.1.1 Face and Eye Landmark Alignment. Accurate landmark alignment is essential for eye feature extraction and head pose estimation. Our work uses one face alignment model to track 48 facial landmarks and two eye alignment models to track the eye landmarks of each eye (see Figure 2). Specifically, we apply the Supervised Descent Method (SDM) [42] to align the facial landmarks. SDM learns descent maps from the training data and uses them to minimize the nonlinear least squares objective without computing the Jacobian or Hessian. It has been shown to achieve robust performance in facial landmarks alignment in the wild [14].

We also train the eye Constrained Local Model (CLM) as in Huang et al.'s [14] work to align the eye landmarks, which are marked in green in Figure 2(b). Unlike the 28-landmark model in Wood et al.'s [40] work, our eye CLM uses only nine landmarks because, in real-use conditions, the low resolution and blurred image does not show clear features; therefore, only points with salient features are tracked to reduce model complexity. To build the eye alignment model, we manually annotated 3,000 eye images from five people (not from our experiment subjects) as in Huang et al. [16].

Since the eye CLM alignment may fail from time to time, it is imperative to validate the eye tracking in real time. To achieve this, we apply a rough but robust tracking model to approximate the iris location, with the assumption that the darkest rectangular area in an eye image is the iris. We define the *eye image* as an enlarged region of the bounding area that encloses the eye contour (e.g., $\boldsymbol{\Theta} = \boldsymbol{c}_7, \boldsymbol{x}_1, \boldsymbol{x}_2, \boldsymbol{c}_8, \boldsymbol{x}_3, \boldsymbol{x}_4$ as shown in Figure 2(b)). In practice, the precision of the landmark alignment can be affected by peripheral noise, such as that from eyelashes or shadows. The aligned landmarks may not be located on the actual eye contour. Therefore, we use the distance of each point from $\boldsymbol{\Theta}$ to represent its probability of being inside/outside the eye and adjust the intensity of the pixels by this probability. Specifically, the intensity I_u of each point $\boldsymbol{u}$ in an eye image is

Fig. 3. Two examples of the eye images and their modified versions for iris detection. In the top row (a) shows an example captured in the dark and (b) in bright light. After the modification based on the probability of being inside/outside the eye contour, the irises in the eye images (in the bottom row) are easier to detect based on intensity.

updated by

$$\begin{cases} I_{\boldsymbol{u}} := I_{\boldsymbol{u}} - a \cdot \text{argmin}_{\boldsymbol{\vartheta} \in \boldsymbol{\Theta}} ||\boldsymbol{u} - \boldsymbol{\vartheta}||, \text{if } \boldsymbol{u} \text{ is inside } \boldsymbol{\Theta} \\ I_{\boldsymbol{u}} := I_{\boldsymbol{u}} + a \cdot \text{argmin}_{\boldsymbol{\vartheta} \in \boldsymbol{\Theta}} ||\boldsymbol{u} - \boldsymbol{\vartheta}|| + b, \text{ otherwise} \end{cases}, \tag{1}$$

where a (=10) controls the impact of probability and b (=50) denotes the confidence of landmark alignment. The parameters a and b are empirically determined and should be adapted to the quality of the face landmark tracking model. In this work, we found a range of values ($a \in [5, 15]$, $b \in [25, 75]$) can produce robust performance for the SDM face tracker.

Figure 3 shows two example eye images before and after this intensity adjustment. We perform min-max normalization to adjust the pixels in the eye image. The result is that the region inside the eye contour is darkened, and the region outside is brightened, depending on the distance of the pixel from the eye contour $\boldsymbol{\Theta}$.

We then search for the darkest rectangle from the integral image [27] of the normalized eye image. The rectangle is determined by the shape of the eye, with its width defined as $0.4||\boldsymbol{c}_7 - \boldsymbol{c}_8||$ and its height as $0.5||\boldsymbol{x}_1 - \boldsymbol{x}_4|| + 0.5||\boldsymbol{x}_2 - \boldsymbol{x}_3||$. Although this approximation of the iris location may not be completely accurate, it is a robust way to determine when the CLM alignment fails. If the eye CLM estimation deviates too much ($>0.4||\boldsymbol{c}_7 - \boldsymbol{c}_8||$) from this approximation, an eye detection will be triggered to reset the CLM alignment. These geometric parameters are adapted from our previous works [14, 16], and they are in accordance with common iris geometries [9].

4.1.2 Iris Center Estimation. The location of the iris center is essential for accurate gaze estimation. However, webcam images can be noisy in *unconstrained* situations due to the low image resolution, reflections on glasses or cornea, shadows, and occlusions from eyelashes. A robust estimation of the iris center generally relies on ellipse-fitting to the contour of the iris or pupil [9]. The conventional edge-based approaches that use gradient and isophote properties [36] are easily affected by visual noises. A recent data-driven method [39] exploited patterns from training data to optimize across all landmarks; however, it may not produce a good fit for landmarks that are located on the limbus. We therefore propose a hybrid method that first performs a rough estimate of the iris based on the eye CLM, followed by an accurate identification of the limbus based on a graph-cut segmentation method.

The GrabCut [29] algorithm is used to segment the iris. GrabCut uses Gaussian mixture models to estimate the color space parameters of the foreground and background. The iris is segmented based on the energy minimization principle, considering both the edge information of the neighboring pixels and the probability of the pixel being the foreground. This generally produces a more

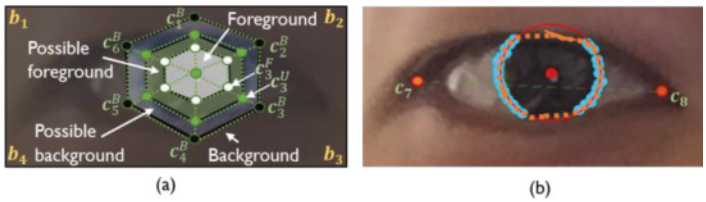


Fig. 4. The procedure of the gaze feature vector extraction based on GrabCut; (a) shows the initial regions of the foreground and background for the GrabCut algorithm, (b) shows the localization of the iris center. The dotted orange shape is the segmentation output of GrabCut, the red circle is fitted to the selected points (marked in blue) on the left and right sides of the limbus contour, and the red dot in the middle is the center of the fitted ellipse (i.e., the iris center).

accurate and robust segmentation of the iris than methods that use only the gray-scale local textual information.

Given the locations of the eye landmarks $\boldsymbol{c} = [\boldsymbol{c}_1, \ldots, \boldsymbol{c}_9]^{\mathrm{T}}$, we can use the following vertices to designate the initial regions of the iris foreground and background:

$$\begin{bmatrix} \boldsymbol{c}_i^F \\ \boldsymbol{c}_i^U \\ \boldsymbol{c}_i^B \end{bmatrix} = \begin{bmatrix} \lambda_F \\ \lambda_U \\ \lambda_B \end{bmatrix} \boldsymbol{c}_i + \begin{bmatrix} 1-\lambda_F \\ 1-\lambda_U \\ 1-\lambda_B \end{bmatrix} \boldsymbol{c}_9, \; i = 1, \ldots, 6, \tag{2}$$

where $\boldsymbol{c}_9$ denotes the iris center estimated by the eye CLM and $\lambda_B(=1.5)$, $\lambda_U(=1.0)$, and $\lambda_F(=0.75)$ are the scale factors determined by the eye geometries [9]. The subscripts denote *B*ackground, *U*ncertain region, and *F*oreground, respectively. We define the initial region of foreground $\boldsymbol{R}^F$, possible foreground $\boldsymbol{R}^{PF}$, possible background $\boldsymbol{R}^{PB}$, and background $\boldsymbol{R}^B$ as

$$\begin{aligned} \boldsymbol{R}^F &= \boldsymbol{P}^F \\ \boldsymbol{R}^{PF} &= \boldsymbol{P}^U - \boldsymbol{P}^F \\ \boldsymbol{R}^{PB} &= \boldsymbol{P}^B - \boldsymbol{P}^U \\ \boldsymbol{R}^B &= \boldsymbol{P}^E - \boldsymbol{P}^B \end{aligned}, \tag{3}$$

where $\boldsymbol{P}^F = \boldsymbol{c}_1^F, \ldots, \boldsymbol{c}_6^F$, $\boldsymbol{P}^U = \boldsymbol{c}_1^U, \ldots, \boldsymbol{c}_6^U$, $\boldsymbol{P}^B = \boldsymbol{c}_1^B, \ldots, \boldsymbol{c}_6^B$, and $\boldsymbol{P}^E = \boldsymbol{b}_1, \ldots, \boldsymbol{b}_4$ are the polygons defined by the given vertices, and $\boldsymbol{b}$ is the vertices of the bounding region of the eye image (see Figure 4(a)). Our experiment uses five iterations in GrabCut, which constrains the computational expense to a reasonable level while generally producing an accurate limbus shape.

Due to occlusion from eyelids and eyelashes, some of the points on the iris contour do not belong to the limbus. Therefore, only the points on the two sides of the iris contour (as shown in Figure 4(b)) are used for ellipse fitting to identify the accurate shape of the limbus. Letting $\boldsymbol{L}$ denote the points on the detected iris contour, we consider the points that satisfy the following condition as belonging to the limbus:

$$\theta_1 < \left| \angle \left(\boldsymbol{l}_i - \boldsymbol{l}_i', \boldsymbol{c}_8 - \boldsymbol{c}_7 \right) \right|_{\boldsymbol{l}_i, \boldsymbol{l}_i' \in \boldsymbol{L}} < \theta_2, \tag{4}$$

where $\angle(\ldots,\ldots)$ calculates the angle between two vectors, $\theta_1(=\pi/4)$ and $\theta_2(=3\pi/4)$ are the angular thresholds depicting the points on the two sides of the contour, and $\boldsymbol{l}_i$ and $\boldsymbol{l}_i'$ are consecutive points on the iris contour $\boldsymbol{L}$ that lie within a small distance (five pixels) of each other (see Figure 4(b)). These parameters are adapted from [15]. We then use Random Sample Consensus (RANSAC) [7] to fit an ellipse to the limbus based on the selected points. The center of the fitted ellipse is considered to be the iris center.

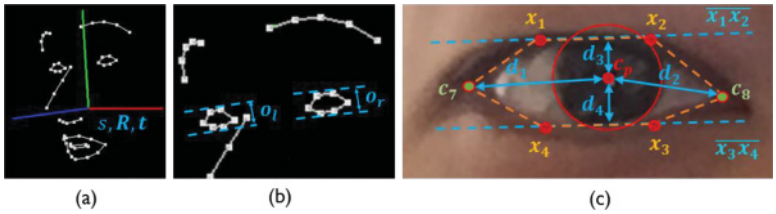


Fig. 5. Illumination of the gaze features used in fast-PACE; (a) shows the 3D shape of the aligned face and the head pose features, including the head scaling s, the three-axis orientation $\boldsymbol{R}$, and the head translation $\boldsymbol{t}$; (b) shows the eye openness features extracted from the 3D normalized shape; (c) shows the horizontal and vertical iris locations measured from the eye image.

4.1.3 Gaze Feature Vector Extraction. Information on the head and eyes is important for gaze estimation [36]. The gaze feature vector used in our study therefore contains the cues of head pose, eye openness, and the locations of the iris centers of both eyes.

We follow previous work [14] to extract the head pose from the aligned facial landmarks (see Figure 5(a)). Procrustes alignment [4] is used to approximate the nonrigid face deformation from the Point Distribution Model (PDM) [5]:

$$\boldsymbol{x}_i = s\boldsymbol{R}\left(\overline{\boldsymbol{x}_i} + \Psi_i \boldsymbol{q}\right) + \boldsymbol{t},\ i \in [1, M], \tag{5}$$

where $\boldsymbol{x}_i$ denotes the 3D coordinate of the ith landmark in the camera coordinate and M(=48) is the number of facial landmarks; $\overline{\boldsymbol{x}_i}$ represents the mean coordinate of the ith landmark; Ψ_i the submatrix of the basis of variation of the ith landmark; and the PDM parameters $\{s, \boldsymbol{R}, \boldsymbol{t}, \boldsymbol{q}\}$ include a head scaling s, a three-axis orientation $\boldsymbol{R}$, a global head translation $\boldsymbol{t}$, and a nonrigid transformation vector $\boldsymbol{q}$.

By removing the rigid geometric variation (s, $\boldsymbol{R}$, and $\boldsymbol{t}$), we measure eye openness from the similarity-normalized face shape. Given the facial landmarks in 3D normalized coordinates (see Figure 5(b)), we define the openness of each eye as

$$o = \alpha \sum_{\boldsymbol{x}_i \in \chi_u,\ \boldsymbol{x}_j \in \chi_l} \|\boldsymbol{x}_i - \boldsymbol{x}_j\|, \tag{6}$$

where χ_u and χ_l represent the sets of landmarks on the upper and lower eyelids, respectively, and α is a constant scale factor that is empirically set to scale the openness of the left and right eyes $\{o_l, o_r\}$ to the range of (0,1).

The location of the iris center in the gaze feature vector is represented by the horizontal location $d_1/(d_1 + d_2)$ and the vertical location $d_3/(d_3 + d_4)$, where d_1, d_2 are the distances between the iris center and the eye corners, and d_3, d_4 are the distances from the iris center to the eyelids $\overline{x_1x_2}$ and $\overline{x_3x_4}$ as shown in Figure 5(c).

An eye feature vector for each eye can then be constructed as

$$\boldsymbol{e} = \left[o, d_1/\left(d_1 + d_2\right), d_3/\left(d_3 + d_4\right)\right]. \tag{7}$$

Concatenating the features from the head and eyes gives us the final gaze feature vector $\boldsymbol{v} = [s, \boldsymbol{r}, \boldsymbol{t}, \boldsymbol{e}_r, \boldsymbol{e}_l]^{\mathrm{T}}$, which contains 12 features, and $\boldsymbol{r}$ is the vector that consists of the diagonal elements of $\boldsymbol{R}$, and $\boldsymbol{e}_r, \boldsymbol{e}_l$ are the eye feature vectors of the right and left eyes, respectively.

4.2 Using Human Behavior to Inform Data Validation

Human gaze patterns can be categorized into four behaviors: *fixation, smooth pursuit, saccade,* and *blink* [9]. A fixation indicates a stationary gaze. Smooth pursuit denotes relatively slow gaze

movements following a moving object, while saccades are eye movements that rapidly direct toward a stationary target.

The behavior-informed validation in fast-PACE identifies moments at which the user's gaze aligns with the cursor/caret when interaction activity is triggered. Based on the knowledge of human gaze patterns and our observations, this most likely occurs during periods of fixation or smooth pursuit. For simplicity, we refer to these periods as stationary period, $\mathcal{S}_s$, and the state of the gaze feature during these periods as being "stationary with small trend" or "stationary." The problem then is to determine $\mathcal{S}_s$ automatically from the webcam data.

4.2.1 Signal Smoothing and Filtering. Based on our previous findings, we focus on the user signals collected in the 3 seconds prior to the interaction event. Linear interpolation is used to resample the signal to 100Hz, giving us 300 samples per feature per interaction event. Since the raw webcam signals contain much high-frequency visual noise, low-pass filtering in the frequency domain is applied to remove the high-frequency temporal jitter from the signal while maintaining the main component without too much distortion. The filtered signal is calculated as

$$\hat{\boldsymbol{f}}_j = \mathcal{F}^{-1}[\mathcal{F}(\boldsymbol{f}_j) * \Omega(\omega)], \tag{8}$$

where $\mathcal{F}$ and $\mathcal{F}^{-1}$ indicate the one-dimensional forward and inverse discrete Fourier transform, respectively, and the filter

$$\Omega(\omega) = \begin{cases} 1, & \omega \leq \omega_h \\ 0, & \text{otherwise} \end{cases}, \tag{9}$$

where ω_h (=2.5Hz, adapted from [14]) is the cutoff frequency.

4.2.2 Extracting a Stationary Feature Vector. To identify the stationary period $\mathcal{S}_s$ and the corresponding estimated feature vector $\widetilde{\boldsymbol{v}}$, our approach adaptively searches for candidate periods occurring close to the point of the event that do not exhibit high signal variations.

To identify these candidate periods, fast-PACE uses a novel adaptive method that searches for a relatively stationary period close to the event. The algorithm is based on that from previous work [14], which uses a threshold to determine if the signal change is small enough to be considered "stationary." However, the previous approach requires a fully fixated gaze and will fail if the gaze is not still, which would be problematic for real-use contexts where the eye is rarely truly stationary. Our approach here relaxes this constraint and is more robust.

Algorithm 1 presents our algorithm. We analyze the changes between consecutive frames in the 3-second gaze window prior to the interaction event. An adaptive threshold $\boldsymbol{\varepsilon}$, which is based on the range of each feature, is used to identify candidate frames in which the feature vectors satisfy the following condition:

$$\prod_{j=1}^{n} H\left(\varepsilon_j^2 - \dot{\hat{f}}_{ij}^2\right) = 1, \tag{10}$$

where $H(x) = (1 + \text{sgn}(x))/2$ is the Heaviside step function and $\dot{\hat{f}}_{ij}$ is the ith frame value of the derivative of $\hat{\boldsymbol{f}}_j$ with respect to the sample time. These frames are considered to be potentially within the stationary period.

After all frames in the window have been tested, a backward search is performed to locate $\mathcal{S}_s$. Linear regression is used to approximate the gaze feature vector $\widetilde{\boldsymbol{v}}$ corresponding to the last moment of $\mathcal{S}_s$, which under our assumptions is also the moment when the user's gaze is most likely to be aligned with the interaction event. If no fixation or smooth pursuit is detected during the gaze window, that interaction event is considered unsuitable for training and is discarded.

ALGORITHM 1: Adaptively Extracting a Stationary Feature Vector

Input: Gaze feature vectors from the 3-second gaze window before the interaction event.

Output: The stationary period $\mathcal{S}_s$ and the feature vector $\tilde{\boldsymbol{v}}$ at the most likely moment of gaze-interaction alignment.

1	Calculate the low-pass frequency-domain filtered signal $\hat{\boldsymbol{f}}_j$ for each gaze feature signal $\boldsymbol{f}_j$ Iteratively search for the stationary period using an incremental threshold
2	$\boldsymbol{\varepsilon} \in [\boldsymbol{r}/1000, \boldsymbol{r}/100]$, where $\boldsymbol{r} \in \Re^{n\times 1}$ is the range of each feature over the gaze window and n is the number of features.
3a.	Initialize the overall stationary period with the frame index $\mathcal{S} = [1, \ldots, m]$, where m is the number of samples.
3b.	Calculate the overall stationary period $\mathcal{S}$ according to Equation (10); set $\mathcal{S}_i$ to 0 if the condition is not satisfied.
3c.	Backward search $\mathcal{S}$ for the first series of consecutive frame indices $\mathcal{S}_s$ whose corresponding duration is longer than the minimum fixation duration (80ms) [9].
3d	**If** t_s, the last moment of $\mathcal{S}_s$, occurs within 0.5 seconds before the event, **then break**. **else** increment $\boldsymbol{\varepsilon}$ by $r/1000$.
	End
4.	Perform line fitting for each feature signal $\hat{\boldsymbol{f}}_j$ in $\mathcal{S}_s$ to approximate the final gaze feature vector $\tilde{\boldsymbol{v}}$, according to Equation (11).

Since features in the stationary state are relatively stable, we apply linear regression to approximate the final gaze feature vector $\tilde{\boldsymbol{v}}$. M-estimator is used to iteratively fit the line using the weighted least-squares algorithm to minimize the distance function

$$\sum_{i\in\mathcal{S}_s} \left(\hat{f}_{ij} - \tilde{f}_{ij}\right)^2/2, \tag{11}$$

where $\tilde{f}_{ij}$ is the fitted value of $\hat{f}_{ij}$. Setting k as the last frame of $\mathcal{S}_s$, the resulting value for the jth gaze feature should be $\tilde{f}_{kj}$. This generates a final feature vector for the classifier, $\tilde{\boldsymbol{v}} = [\tilde{f}_{k1}, \ldots, \tilde{f}_{kn}]^{\mathrm{T}}$.

4.2.3 Data-Driven Validation. Behavior-informed validation looks for the stationary period corresponding to the interaction event. However, there is a possibility that even though the user's gaze is fixated on something, the gaze point may not be anywhere near the location of the interaction event—for example, when the user is watching a movie with the mouse pointer poised over the "pause" button or typing a chat message while reading the previous incoming responses. To accommodate these types of interactions, fast-PACE uses data-driven validation as an additional layer of validation to determine the goodness of the feature vector $\tilde{\boldsymbol{v}}$ and the corresponding assumed interaction-informed gaze point $\boldsymbol{g}^{\text{event}}$ based on previously validated data.

We use support vector regression [31] to build the gaze estimation models for both x- and y-coordinates. The gaze feature vectors $\tilde{\boldsymbol{v}}$ are used as features, and the corresponding interaction points $\boldsymbol{g}^{\text{event}}$ are considered as the "truth." A second-order polynomial kernel is used to allow for nonlinear learning, and the complexity parameter is set to 1. The initial model is trained on the first 1,000 interaction instances in which the moment of the interaction occurs within a fixation period. We then learn the mapping function from gaze feature vector to gaze point $\mathrm{h}^{\text{target}} : \boldsymbol{v} \rightarrow \boldsymbol{g}$.

When the most recently collected feature vector $\tilde{\boldsymbol{v}}$ is passed through the gaze model, it outputs a webcam-estimated gaze point $\boldsymbol{g}$. If $\|\boldsymbol{g} - \boldsymbol{g}^{\text{event}}\| \leq \lambda_s$, where λ_s(=1/12 of the screen diagonal length in our work) is the error tolerance and $\boldsymbol{g}^{\text{event}}$ is the interaction-informed gaze point, also the location of the corresponding interaction event, the instance $[\tilde{\boldsymbol{v}}, \boldsymbol{g}^{\text{event}}]$ is considered validated and can be used as target training data. For efficiency, the gaze model is updated in a batch mode; upon the collection of 1,000 valid instances of new target training data, all the validated data will be used to retrain the user-specific gaze models or the user-adaptive models taken together with the data from the source subjects. To make full use of all potential data, instances that fail the current validation are also retained and re-evaluated after each update of the gaze model.

In summary, fast-PACE uses a dual-level validation. First, by looking for the stationary gaze features corresponding to an interaction event, the behavior-informed validation uses knowledge of gaze movement patterns and user interaction to identify the moment when user gaze and interaction point are most closely aligned. The data-driven validation further applies prior knowledge of the particular user to account for user individualities and contextual differences.

5 GAZE TRANSFER FROM SOURCE CLASSIFIERS

The dual-level validation identifies noisy interactional cues for gaze learning. Given the gaze feature vector $\boldsymbol{v}$, the user-specific component of our method (PACE) uses the regression model $\mathrm{h}^{\text{target}}$ to continuously estimate the gaze point:

$$\boldsymbol{g}^{\text{PACE}} = \mathrm{h}^{\text{target}}(\boldsymbol{v}) . \tag{12}$$

However, in real-use scenarios, collecting enough validated data is slow, and the accuracy of the gaze model would remain suboptimal for a long period of time. Fast-PACE therefore applies transfer learning to best exploit limited data from the target user and make use of the source data that can be prepared in advance. In other words, fast-PACE makes proper use of gaze transfer, particularly when there is insufficient data to achieve a good user-specific model, to build a *personalized* model for the target user.

5.1 Building the Gaze Ensemble Classifier

Specifically, given a source dataset $\mathcal{D}^{\text{source}}$ with N subjects and the validated data of the target user $\mathcal{D}^{\text{target}} = \{\tilde{\boldsymbol{v}}_i, \boldsymbol{g}_i^{\text{event}}\}_{i=1}^{n^{\text{target}}}$ that contains n^{target} instances, we train an ensemble classifier for the target user which aggregates the source classifiers $\{\mathrm{h}_j^{\text{source}}\}_{j=1}^{N}$ and the target classifier $\mathrm{h}^{\text{target}}$ based on their performances on the target data $\mathcal{D}^{\text{target}}$.

It is not feasible to directly use the raw source classifiers due to the interpersonal differences between source subjects and the target user. Nevertheless, we observe that even a source classifier with a large mean absolute error can yield a high correlation coefficient between estimations and true gazes. This is in line with findings from pursuit-based gaze interaction [38]. We therefore use the correlation coefficient as the performance metric and identify similar sources with correlations larger than a given threshold λ_c(=0.7), which is empirically set to ensure a high correlation between estimations of target and source classifier. To compensate for the impact of interpersonal differences, we introduce an adaptation mapping Φ to adjust the estimation of the source classifier for the target user. The algorithm of constructing the personalized gaze model is summarized in Algorithm 2.

A general form of the ensemble classifier that leverages the source knowledge can be written as

$$g^{\text{transfer}} = \sum_{j \leq |\mathcal{H}|} w_j \Phi_j \left(g_j^{\text{weak}}, v \right) , \tag{13}$$

ALGORITHM 2: Learning the Ensemble Classifier for Fast-PACE

Input: The source set $\mathcal{D}^{\text{source}}$ and the validated data of the target user $\mathcal{D}^{\text{target}}$

Output: A set of weak classifiers $\mathcal{H}$, the corresponding weights $\mathcal{W}$, and the adaptation mappings Φ.

Phase I. Learning the set of weak classifiers $\mathcal{H}$ from all the N source subsets $\mathcal{D}^{\text{source}} = \{\mathcal{D}_1^{\text{source}}, \ldots, \mathcal{D}_N^{\text{source}}\}$ and the target set $\mathcal{D}^{\text{target}}$

for each data subset $j = 1$ to N

1a Learn a weak classifier $\mathrm{h}_j^{\text{source}}$ on $\mathcal{D}_j^{\text{source}}$

1b Update the weak classifier set $\mathcal{H} \leftarrow \mathcal{H} \cup \{\mathrm{h}_j^{\text{source}}\}$

End

2 Learn the target classifier from $\mathcal{D}^{\text{target}}$ and update the weak classifier set $\mathcal{H} \leftarrow \mathcal{H} \cup \{\mathrm{h}^{\text{target}}\}$

Phase II. Learning an ensemble classifier on target training set $\mathcal{D}^{\text{target}}$ for each weak classifier $\mathrm{h}_j \in \mathcal{H}$

3a Evaluate h_j on $\mathcal{D}^{\text{target}}$, calculate correlation $w_j = \text{get_corr}(\{\mathrm{h}_j(\tilde{\boldsymbol{v}}_i)\}_{i=1}^{n^{\text{target}}}, \{\boldsymbol{g}_i^{\text{event}}\}_{i=1}^{n^{\text{target}}})$

3b **If** $w_j < \lambda_c$ **then** set w_j to 0.

3c Learn the adaptation mapping Φ_j from the output of weak classifier $\boldsymbol{g}_j^{\text{weak}}$, gaze feature vector $\tilde{\boldsymbol{v}}$, and the interaction informed point $\boldsymbol{g}^{\text{event}}$

3d Update weights $\mathcal{W} \leftarrow \mathcal{W} \cup \{w_j\}$ and adaptation mapping $\Phi \leftarrow \Phi \cup \{\Phi_j\}$

End

4 Normalize each element of the weight vector $w_j = w_j / \sum_i w_i$

return $\{\mathcal{H}, \mathcal{W}, \Phi\}$

where $\mathcal{H}$ is the set of weak classifiers, $|\mathcal{H}|$ is the number of weak classifiers in $\mathcal{H}$, w_j is the weight, and $\boldsymbol{g}_j^{\text{weak}}$ is the estimation of the jth weak classifier. The final gaze estimation is the weighted aggregation of the predictions given by all the weak classifiers in $\mathcal{H}$ with non-zero weights.

The performance of the personalized model constructed with gaze transfer depends on the amount of available target data and its similarity to the sources [2]. If the source data are highly dissimilar to the target data, the user-specific model from PACE might well outperform the personalized model, even with fewer data points. Fast-PACE therefore uses the interaction location corresponding to the gaze point as the ground truth annotation for that data point. The estimated location of the gaze point from the personalized and user-specific models is periodically compared against this ground truth to evaluate the performance of gaze learning and personalization. This allows fast-PACE to dynamically switch between the user-specific estimation $\boldsymbol{g}^{\text{PACE}}$ and the personalized estimation $\boldsymbol{g}^{\text{transfer}}$ based on their evaluated performances in real time; that is,

$$\boldsymbol{g}^{\text{fast-PACE}} = \begin{cases} \boldsymbol{g}^{\text{transfer}}, & \text{if transfer outperforms} \\ \boldsymbol{g}^{\text{PACE}} & \text{, otherwise} \end{cases}, \tag{14}$$

Compared with the feature-based transfer learning approach, the user adaptation method used in this study is much faster and thus more suitable for practical use because our method only needs to reorganize and reweigh the source classifiers without performing any additional retraining. In contrast, as the amount of target data increases, the feature-based and parameter-based transfer learning approaches require target model updates using all source samples, which can be computationally expensive. If the source dataset is large, this online learning process may result in periodical heavy loads on end-user personal devices.

5.2 Adapting the Estimations of Source Classifiers

The knowledge adaptation for $\boldsymbol{g}^{\text{transfer}}$ depends on the adaptation mapping Φ. This section describes two representations for gaze transfer. The evaluation section will then present a further comparison.

Inspired by the work of Lu et al. [22] that decomposes head pose variations from eye appearance and that of Sugano et al. [35] that considers head pose for gaze correction, we explore a Shared Pose Transfer Model (SPTM)which that bridges the gaze transfer among different individuals by gaze estimation under the same head pose (e.g., frontal view). Specifically, we divide the data of each subject into two parts, depending on whether or not he or she faces the screen straight-on. We then construct separate classifiers on each part of the data from individuals; namely, $\mathrm{f}: \boldsymbol{v} \rightarrow \boldsymbol{g}$, when $\boldsymbol{v}$ is describing a frontal view, or $\mathrm{nf}: \boldsymbol{v} \rightarrow \boldsymbol{g}$, when $\boldsymbol{v}$ is under a nonfrontal view.

The gaze estimation Equation [13] for SPTM can be written as

$$\boldsymbol{g}^{\text{SPTM}} = \sum_{j \leq |\mathcal{H}|} w_j \Phi_j \left(\mathrm{nf}_j^{\text{source}}(\boldsymbol{v}), \mathrm{f}_j^{\text{source}}(\boldsymbol{v}), \mathrm{f}^{\text{target}}(\boldsymbol{v}), \boldsymbol{v}\right), \tag{15}$$

where $\mathrm{f}_j^{\text{source}}$ and $\mathrm{f}^{\text{target}}$ are the classifiers built on the *frontal data* of the jth source subject and the target user, respectively; and $\mathrm{nf}_j^{\text{source}}$ denotes the source classifier trained on the *nonfrontal data*. By doing so, we hypothesize that the ensemble classifier can learn the impact of head pose variation from $\{\mathrm{f}^{\text{source}}(\boldsymbol{v}), \mathrm{nf}^{\text{source}}(\boldsymbol{v})\}$ and measure the interpersonal difference from $\{\mathrm{f}^{\text{source}}(\boldsymbol{v}), \mathrm{f}^{\text{target}}(\boldsymbol{v})\}$.

On the other hand, inspired by Occam's razor, we propose a Direct Transfer Model (DTM), which adapts the estimation of the source classifiers without explicitly considering head pose. The gaze estimation of DTM can be written as

$$\boldsymbol{g}^{\text{DTM}} = \sum_{\mathrm{h}_j \in \mathcal{H}} w_j \Phi_j(\mathrm{h}_j(\boldsymbol{v}), \boldsymbol{v}), \tag{16}$$

where h_j is the weak classifier constructed according to Algorithm 2. Compared with SPTM, DTM does not require separate data collection of the frontal view. It relaxes the constraint imposed on training data and coincides with our objective of implicit gaze learning from interactions.

In our experiment, we use support vector regression [31] to learn the source classifiers as well as the functions for the adaptation mapping. Since the adaptation is a 2D calibration of the gaze estimation, we use a linear kernel for Φ; the gaze estimation classifiers use the same configuration (with a second-order polynomial kernel) as $\mathrm{h}^{\text{target}}$ in the data-driven validation.

6 EVALUATION OF FAST-PACE

This section presents an evaluation of fast-PACE. More specifically, we intend to answer three important questions pertinent to gaze learning from noisy and limited interactional cues: (i) *Accuracy of iris center localization in real use:* Can fast-PACE accurately locate the iris center from the webcam images? (ii) *Gaze-interaction alignment:* Can fast-PACE correctly identify the moment when people gaze at where they click or type more precisely? (iii) *Effectiveness of knowledge adaptation:* Can fast-PACE rapidly construct an accurate gaze model from a modest amount of personal interaction data?

6.1 Evaluating Iris Center Localization

The accurate localization of iris center is the foundation of the geometric-based gaze estimation approach adopted in fast-PACE. We first evaluate fast-PACE by comparing its accuracy of iris localization with state-of-the-art on two public datasets, BioID [19] and Gi4E [28].

BioID consists of 1,521 grayscale images with a resolution of 384 × 286 pixels. These images are collected from 23 different subjects. Gi4E contains 1,339 color images from 103 subjects. The

Table 2. The Performance of Our Approach, Compared with State-of-the-Art Normalized Error Scores on the BioID Dataset

Methods	$e \leq 0.05$	$e \leq 0.1$	$e \leq 0.25$
Fast-PACE (ours)	**94.3%**	99.3%	**100.0%**
[54]	93.8%	**99.8%**	99.9%
[50]	91.7%	98.2%	99.9%
[24]	89.9%	97.1%	99.7%
[3]	87.2%	93.4%	≥93.4%
[37]	86.1%	91.7%	97.9%
[8]	85.1%	94.3%	≥98.1%

Table 3. The Performance of Our Approach, Compared with State-of-the-Art Normalized Error Scores on the Gi4E Dataset

Methods	$e \leq 0.05$	$e \leq 0.1$	$e \leq 0.2$
Fast-PACE (ours)	**99.7%**	**100.0%**	**100.0%**
[8]	89.3%	92.3%	94.2%
[1]	81.4%	89.3%	89.9%

images have a resolution of 800×600 pixels. We use the manually labeled 2D iris centers provided by the dataset as ground truth for evaluation.

Following the previous practice [19], we define the normalized error as

$$e = \frac{\max\{d_L, d_R\}}{d_E}, \tag{17}$$

where $d_L(d_R)$ represents the Euclidean distance between the location of the estimated iris center and the ground truth location of the left (right) eye, and d_E indicates the Euclidean distance between the iris centers of the ground truth. The evaluation metric is the rate at which the iris is detected within the normalized error. A normalized error of $e \leq 0.25$ roughly corresponds to half the distance between the two corners of an eye, $e \leq 0.1$ corresponds to the diameter of the iris, and $e \leq 0.05$ corresponds to the range of the cornea [36].

Tables 2 and 3 show the performance of fast-PACE compared with state-of-the-art algorithms on BioID and Gi4E, respectively. It can be seen that fast-PACE generally outperforms state of the art on both datasets. Since our gaze estimation relies on precise iris center localization, the accuracy within the smallest normalized error ($e \leq 0.05$) is of the most interest to us. It is encouraging that fast-PACE achieves the highest detection rate ($e \leq 0.05$), with 94.3% accuracy on BioID and 99.7% on Gi4E.

To better understand the performance and the causes of errors, we inspect some examples of success and failure scenarios (see Figure 6). Row (a) presents two successful examples on each dataset. The results are satisfactory. Fast-PACE can accurately extract the iris centers even when the iris is partially occluded by eyelids or affected by reflections on glasses. We can also see that the iris contour can be correctly estimated even when the users are not gazing at the screen straight-on.

Inspecting the failure cases, our results reveal that annotation mistakes are one important cause of error on BioID, notably when eyes are not fully opened and when there are strong reflections on glasses, as shown in Figure 6 rows (b) and (c).

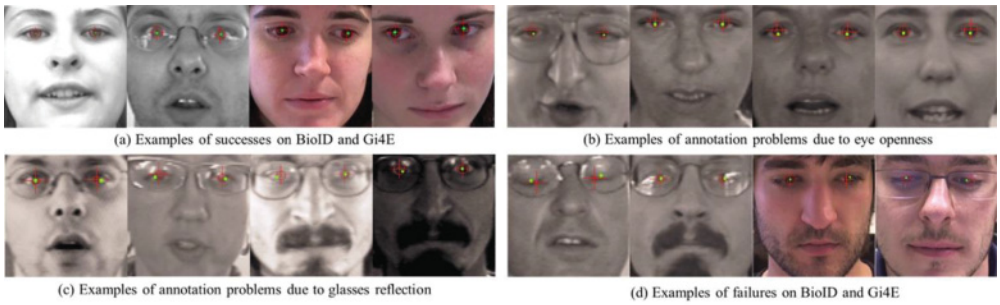


Fig. 6. Examples of iris center localization based on fast-PACE on BioID (Jesorsky et al. 2001) and Gi4E (Ponz et al. 2012). The green dot denotes the ground truth annotation of the iris location, the red circle indicates the fitted ellipse of limbus, and the red cross shows the estimated center of the iris. Row (a) shows examples of successful localization. The grayscale examples are from BioID and the color examples from Gi4E. Row (b) shows examples of annotation mistakes when the eyes are not fully opened. Row (c) shows examples of annotation mistakes due to reflections on eye glasses. Row (d) gives examples of localization failures.

Although these factors also affect the accuracy of fast-PACE to a certain degree (see Figure 6 row (d)), the errors caused by eye openness often are due to eye blinks or the gaze looking away from the screen. The two-layer validation of fast-PACE can effectively identify such circumstances and inform applications in real use.

To conclude, even though the degree of eye openness and reflections on glasses may affect the localization of iris center, fast-PACE can achieve high accuracy in identifying the iris center for an absolute majority of cases. Compared with state of the art, fast-PACE produces better estimation results under small normalized error ($\leq$0.05). It therefore shows promising potential for gaze estimation in the wild.

6.2 Evaluating Validation Mechanism for Interaction Data

Tracking and learning gaze location relies on accurate localization of the iris center. In addition, the availability of adequate and valid training data is equally important in constructing a well-performing gaze model. In this section, we evaluate the validation mechanism adopted in fast-PACE. In other words, we investigate the identification of temporal and spatial alignment between gaze and interaction.

6.2.1 Collecting an Interaction Dataset with Predefined Tasks. Three experiments were designed to generate the necessary interaction behaviors for alignment evaluation, including mouse-click, mouse-drag, and keypress actions. The first experiment requires the subject to click on targets of different shapes and sizes, including (i) long slim targets, (ii) small targets, and (iii) large targets. A list of academic papers with long titles ($\geq$3/4 of the screen width) was prepared in advance. The subjects were asked to search for each paper on Google Scholar and to click on the hyperlinks for the title and authors for each paper. Since Google abbreviates authors' first names, author hyperlinks are usually short and button-like (small targets), and the hyperlinks for paper titles give long slim targets. The large targets were modeled via photos in Flickr, for which the subjects were asked to search and click on those photos that they found interesting. The width of the photos occupied around one-third of the screen. The second experiment requires subjects to select sentences from a given PDF document by dragging the mouse. For the third experiment, subjects were asked to type a short paragraph into a text file. We constrained ourselves with only letter

keys ("a"–"z," "A"–"Z," spaces) because most people, even those who can touch-type, are actually only able to touch type the letter keys and don't perform as well on number or function keys.

We used the Tobii EyeX tracker to acquire the "ground truth" of the user's gaze point. The Tobii tracker adopts infrared technology with 60Hz tracking frequency and achieves an accuracy within 1 degree of visual angle (corresponding to 30–50 pixels on our 22-inch monitor at 1,680 × 1,050 resolution and a reading distance ≈500–800mm), which can be considered state of the art. The video stream processed by fast-PACE is captured from a standard webcam mounted on the top center of the monitor.

We recruited 31 subjects (16 female; aged 20–30 yrs, M = 25.1, SD = 2.5) for this study. The subjects were university students and staff. Among the subjects, 24 are able to touch-type, at least using letter keys. Subjects were asked to work naturally, which meant free movement of head and body. Each of the subjects was required to generate at least 50 instances of each interaction activity. The occurrence of a clicking event was defined as the press of the left mouse button, dragging events as the release of the left mouse button, and keypresses as the depression of a letter key. Each interaction event triggered a screenshot that was saved for data validation and event classification. In total, we collected 1915 clicking events on long slim targets, 2,344 on small targets, and 1,955 on large targets; 2,029 dragging events; and 4,863 typing events.

6.2.2 Evaluating the Identification of Gaze-Interaction Alignment. For each interaction event, we compare its location $\boldsymbol{g}^{\text{event}}$ with the Tobii-measured gaze coordinates $\boldsymbol{g}^{\text{tobii}}$ at the moment identified by the behavior-informed validation mechanism as the point of closest alignment. The error of our method, $error_{fast-PACE} = \|\boldsymbol{g}^{\text{tobii}} - \boldsymbol{g}^{\text{event}}\|$, is calculated as the displacement (Euclidean distance) between $\boldsymbol{g}^{\text{tobii}}$ and $\boldsymbol{g}^{\text{event}}$. As a baseline comparison, we make the naïve assumption and choose the moment of the interaction event as the moment of best alignment, as in previous work [33]. The resultant error at moment i (of interaction) is denoted $error_{naive} = \|\boldsymbol{g}_i^{\text{tobii}} - \boldsymbol{g}^{\text{event}}\|$. In order to determine a bound on the error, we compute $error_{min}$ as the minimum possible error that one could achieve if one could somehow magically know the exact moment of the best gaze-interaction alignment in the 3-second window. Stepping up human intervention, one could also discard the obvious errors in terms of outliers—those points at a displacement larger than λ_s (1/12 of the screen diagonal length ≈240 pixels)—to compute a tighter bound on the error as $error_{minf}$. Those outliers often arise naturally when the user is clearly not looking at the interaction point; for example, when he or she is looking for a key on the keyboard.

Figure 7 depicts the performance achieved by the different approaches on the data described in this experiment, namely, the baseline naïve approach, our approach, and the conceptual bounds with/without data validation. The color bars indicate the displacement values, and the circles denote the percentage of data that passes the validation process after outliers are removed.

The results are encouraging. As expected, $error_{min}$ (dark green) can be very small, with an average of 41.1 pixels over all different patterns of mouse interactions. The exception is with keyboard events, where $error_{min}$ is observed to be much larger at 299.7 pixels. This is typically caused by the instances in which the user was not looking at what was being typed at all—the gaze was either wandering about the screen, focusing on the keyboard, or even looking away for idea formulation. When those outliers are discarded, the keyboard event error $error_{minf}$ (light green) decreases rapidly to 46.7 pixels. Discarding outliers for all events brings the average error down to 27.8 pixels. This corroborates the objective of our data-driven validation that removes estimations with large deviations. However, $error_{minf}$ provides a bound on the best achievable result, which is extremely difficult to achieve in practice. Nevertheless, it can be observed that our approach in general yields a close enough performance toward this best scenario.

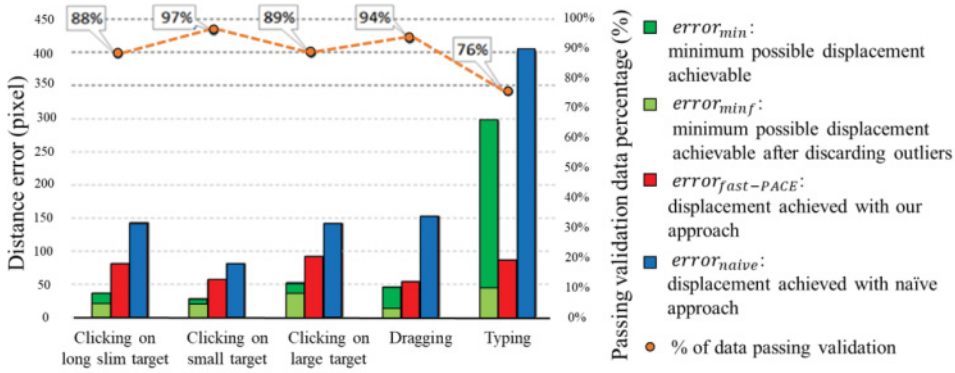


Fig. 7. Displacement between gaze points identified with different approaches and location of corresponding interaction events.

It is also interesting to consider the amount of data retained after the two-layer validation process. In Figure 7, it can be observed that 88.8% of the data are retained on average. In particular, more than 90% of mouse-related interaction data are retained, whereas only 76% of keyboard data are retained. This is due to a relatively large number of outliers generated when subjects were looking away from the typing point. Incidentally, both the unfiltered $error_{min}$ and $error_{naive}$ are on the high side for typing interactions, especially the naïve approach (401.8 pixels). This phenomenon is perhaps due to the fact that subjects do not need to look at the screen to know what they are typing, in contrast to mouse clicks that require looking at the screen for placement. Nonetheless, our method is still able to successfully identify these problematic data points during typing interactions to achieve a small $error_{fast-PACE}$, which is close to that of mouse events. This is promising because keypress events are usually more numerous than mouse events, and thus it makes sense to find a means to include them as interaction-informed data.

Our results suggest that the fast-PACE validation mechanism can effectively and precisely identify the reliable interaction-informed data to be included in the gaze model, significantly outperforming existing methods based on the conventional assumption that users gaze where they click or press.

6.3 Evaluating Fast-PACE in Real-Use Contexts

This section evaluates the proposed method for gaze modeling in real-use contexts. We first present the user data collection mechanism in our experiments, followed by an in-depth evaluation of our gaze transfer technique. Finally, we summarize fast-PACE and compare it with PACE and other state-of-the-art techniques.

6.3.1 Collecting a Real-Use Dataset. To evaluate the accuracy and effectiveness of fast-PACE in real-use contexts, we recruited nine subjects (university students, five female; aged 20–33; M = 27.4, SD = 3.6) for a focused study. Subjects were asked to choose at least three of the following tasks for data collection: browsing websites, coding in Visual Studio, writing in Notepad, creating a figure using Microsoft Paint, and playing a shooting game (the House of the Dead). These tasks were chosen to cover a diverse range of common user interaction activities and applications and to generate diverse interaction types. For example, some of the tasks will involve relatively dense keypresses while others contain mainly mouse events, like clicking and dragging.

The data we collected from these tasks are used as training data for gaze learning. Each mouse click and press of a letter key logs the gaze feature data from the preceding 3 seconds. Subjects were free to adjust their head pose, body posture, and chair position/height. For evaluation, we followed the protocol from Zhang et al. [51], which collects a test dataset by requiring the subjects to gaze at an animated calibration point that moves around the screen at a constant speed and removing eye blinks or saccades. We collected the test data every 30 minutes during interactions in order to capture head pose variations. For each subject, we conducted 10 sessions of test data collection, each of which lasted around 1 minute and generated around 1,000 instances. In total, we collected 364,219 instances (click/type) for all subjects over the course of 37 days (i.e., on average, around 30,000 training instances and 10,000 test instances per subject).

The experiments were run on an i7-2600MHz PC with 4GB RAM, a 22-inch monitor, and a standard off-the-shelf webcam capable of recording at 30fps. Running fast-PACE on this setup achieves a frame rate of 22fps, and performing a model update with 1,000 data points takes less than 300ms. Approximated by our face tracker, in our test dataset, the head-to-monitor distance ranges from 445 to 864mm (M = 616mm, SD = 63.4mm); and head-to-camera pitch ranges from − 17.2∼40.1 degrees (M = 17.9 degrees, SD = 5.6 degrees).

To constrain the computation of model learning and update, we apply a similar resampling technique as Sugano et al. [34] on the training set to retain 30% of the instances, which are spread uniformly across the screen. To evaluate the gaze estimation performance on the test set, we present the correlation coefficient and the visual error between the outputs of our gaze regression models and the gaze points. Visual error in degrees is calculated using the approach in Sugano et al [33].

6.3.2 Evaluating Gaze Transfer. Section 5 proposed a gaze transfer learning technique based on current gaze and head pose states. Equation (13) shows how our adaptation takes the 12-dimensional gaze feature vector $\boldsymbol{v}$ as input, which contains the head pose and eye geometric information. However, these features may not all be useful for adaptation. We therefore perform an evaluation using different input features to seek the optimal form of adaptation. We define the transfer mechanisms with different inputs as follows:

- $\mathrm{SPTM}_0, \mathrm{DTM}_0$: Shared Pose Transfer Model (SPTM) and Direct Transfer Model (DTM) that consider neither eye nor head features.
- $\mathrm{SPTM}_{\mathrm{head}}, \mathrm{DTM}_{\mathrm{head}}$: Models that consider head features.
- $\mathrm{SPTM}_{\mathrm{eye}}, \mathrm{DTM}_{\mathrm{eye}}$: Models that consider eye features.
- $\mathrm{SPTM}_{\mathrm{head+eye}}, \mathrm{DTM}_{\mathrm{head+eye}}$: Models that consider both head and eye features.

As the amount of target data is crucial to the performance of gaze transfer learning, we plot 10 performance data points obtained with different amounts of target samples, each of which consists of 1,000 incremental instances.

Figure 8 shows the performance comparison of SPTM with and without using eye and head features for adaptation. We see that input features can significantly affect gaze transfer. Generally, performance of gaze transfer increases with the number of target samples. Unexpectedly, $\mathrm{SPTM}_{\mathrm{head}}$ does not lead to any improvement and may actually result in a decreased performance even when there are moderately sufficient data instances (∼7,000 to 8,000). Nevertheless, it is encouraging that, given sufficient target user data, visual errors achieved by the other three SPTMs gradually converge at around 2.7 degrees, which is a far more important/significant performance metric than the correlation coefficient. More importantly, with limited target data (the first three epochs), adaptations without using head features clearly outperform their counterparts. Inspecting the highest initial correlation (SPTM_0: 0.82), the best-performing SPTM does not even need to

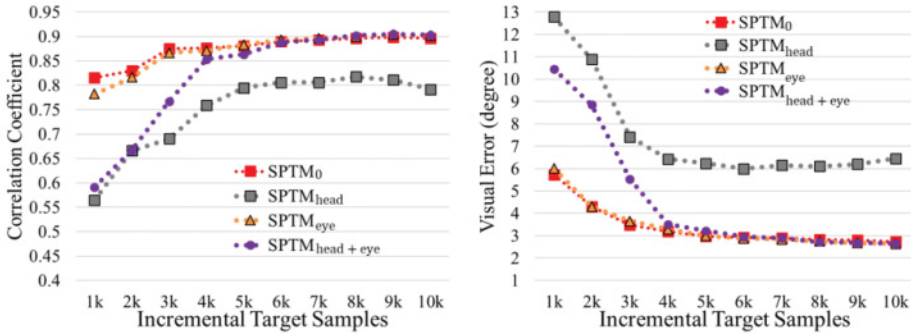


Fig. 8. Performance comparison of SPTM with different input features and on increasing size of target training dataset. Each epoch consists of 1,000 incremental instances. Adaptations without head features give better results.

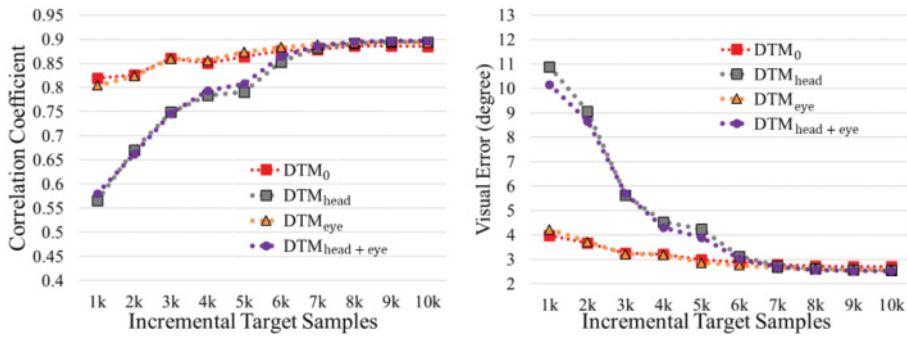


Fig. 9. Performance comparison of DTM with different input features and on increasing size of target training dataset. Similar to SPTM, DTMs without head features give better results.

rely on the current eye and head states, which is in line with the law of parsimony (i.e., Occam's razor).

Figure 9 presents the performance comparison of DTM with different input features and on increasing size of target samples. DTM presents more promising performance than SPTM as it gives higher performance in the initial stage with limited target data (DTM_0: 3.98 degrees; SPTM_0: 5.71 degrees). Furthermore, all DTMs show steadily increasing performance with incremental target samples. With respect to the adaptation inputs, a consistent pattern emerges: Taking head features into consideration seems to deteriorate the adaptation process.

Our results suggest that a straightforward correction of the gaze estimation of the source classifier is preferable. In this sense, the gaze estimation of SPTM (Equation (15)) and DTM (Equation (16)) can be rewritten as

$$\begin{aligned} \hat{\boldsymbol{g}}^{\text{SPTM}} &= \sum_{j \leq |\mathcal{H}|} w_j \Phi_j \left(\text{nf}_j^{\text{source}}(\boldsymbol{v}), \text{f}_j^{\text{source}}(\boldsymbol{v}), \text{f}^{\text{target}}(\boldsymbol{v}) \right) \\ \hat{\boldsymbol{g}}^{\text{DTM}} &= \sum_{\text{h}_j \in \mathcal{H}} w_j \Phi_j(\text{h}_j(\boldsymbol{v})) \end{aligned} . \tag{18}$$

This is the form of the SPTM and DTM (i.e., SPTM_0 and DTM_0) that will be used in the following evaluation.

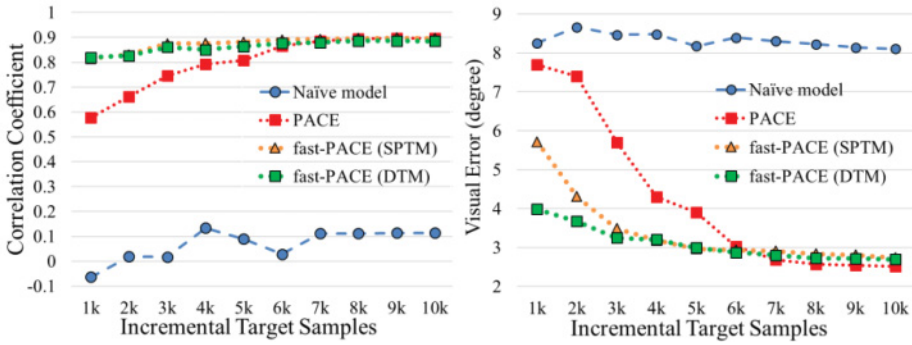


Fig. 10. Comparison of user-specific gaze model with (i) naïve assumption, (ii) validation mechanism (i.e., PACE), (iii) fast-PACE (SPTM), and (iv) fast-PACE (DTM). The change in performance, as measured in both Correlation and Visual Error, as the amount of target data increases, shows that data validation ensures learning effectiveness, and fast-PACE (DTM) successfully reduces the target samples required for a well-performing gaze model.

6.3.3 Comparing Fast-PACE with PACE. To demonstrate the effectiveness of fast-PACE (i.e., PACE + SPTM/DTM), this section discusses the performance comparison among different gaze learning paradigms, including gaze models with (i) the naïve assumption that users gaze where they click/type, (ii) validation mechanism (i.e., PACE; 13), (iii) fast-PACE (SPTM), and (iv) fast-PACE (DTM).

Figure 10 shows the correlation and visual error of these models. As the number of target samples increases, the performance of the naïve model (marked with blue dots) fluctuates considerably: the visual error hovers around 8 degrees, and its correlation never increases beyond 0.2. This shows that a large proportion of the data generated by interaction events does not satisfy the gaze-interaction consistency, and direct learning from such "inconsistent/dirty" data will not facilitate the building of an accurate model.

If a simple validation mechanism is added, without a transfer learning mechanism like PACE (red squares), the performance improves monotonically as additional training data are provided. The average correlation (on x- and y-coordinates) reaches an impressive value of 0.90. The visual error of PACE drops steadily to 2.52 degrees, which represents the best performance across all different models. This is a further piece of evidence that shows our validation mechanism is effective as well as necessary for collecting interaction-informed data in real-use situations for gaze learning.

Even though PACE can successfully estimate the gaze location when there are sufficient training samples, the initial visual error, when only a limited amount of target data is available, is only marginally better than the naïve method. As shown in Figure 10, a well-performing PACE model (e.g., visual error $\leq$3 degrees) needs around 6,000 validated training samples, which may take hours or days to collect.

However, it is encouraging that gaze transfer techniques give promising results with a much more modest amount of target samples. Fast-PACE (SPTM) (orange triangles) and PACE (DTM) (green squares) present similar correlations with an initial value of 0.82, which is much higher than that of PACE (0.58). In terms of visual error, fast-PACE (DTM) leads to a markedly higher accuracy with an average error of 3.98 degrees, which is 1.73 degrees and 3.72 degrees less than those of fast-PACE (SPTM) and PACE, respectively. This indicates that the DTM transfer mechanism can significantly improve PACE by adapting from the source knowledge for personalized eye estimation especially when the available target data are limited.

Table 4. The Performance of Our Approach Compared with State-of-the-Art Webcam-Based Models that Allow Free Head Motion

Methods	Error	Calibration	Data Required / Method Used
Fast-PACE (initial)	**3.98°**	Implicit	Mouse/keyboard interactions
Fast-PACE (final)	**2.52°**	Implicit	Mouse/keyboard interactions
[41]	~7°	NA	Based on trigonometric
[33]	4°–5°	Implicit	Click
[34]	~3°	Implicit	Click
[22]	2°–3°	Explicit	Video
[23]	2°–3°	Explicit	Image synthesis
[37]	3°–5°	Explicit	Calibration points
[3]	2°–3°	Explicit	Calibration points

A close scrutiny of the performance changes of PACE and fast-PACE (DTM) shows an interesting finding. Although fast-PACE (DTM) outperforms PACE with insufficient target data, PACE learning from only the user-specific data can eventually achieve a higher accuracy. The performances cross at around 6,000–7,000 target samples. Since our method keeps track of the interaction-informed data, we can therefore periodically evaluate the performances of PACE and fast-PACE (DTM) and dynamically select the better one. In this sense, the overall performance of fast-PACE with different target samples can be approximated by the best performance between PACE and fast-PACE (DTM).

Table 4 summarizes the performance achieved by fast-PACE compared against similar state-of-the-art methods that rely on webcam signals. The performance is measured as the average error over all collected instances and subjects. It is encouraging to see that our method achieves an average visual error of 2.52 degrees, which is favorably comparable to state-of-the-art approaches that require explicit calibration. This is a promising result, given that fast-PACE (i) requires no necessary explicit calibration and will automatically update itself to account for changes in light and posture variations, even across multiple sessions spanning over multiple days; (ii) uses conventional off-the-shelf equipment that is commonplace in work environments; and (iii) has been tested using real applications and activities.

7 CONCLUSION AND FUTURE WORK

This article describes fast-PACE, a fast approach to the Personalized, Auto-Calibrating Eye-tracking system that can be integrated into standard interactive computing systems. Through an in-depth study of gaze-interaction consistency, we show that the assumption that people gaze where they click or type does not always hold in real-use human–computer interaction contexts. Based on the results of this study, we then develop a novel approach that automatically identifies the moment of best gaze-interaction alignment and a further data validation mechanism that accounts for user differences and context variations. In addition, to meet the practical need of gaze-aware applications, we propose the gaze transfer technique to address the lengthy data acquisition problem by applying interactional cues for gaze learning.

Experimental evaluations demonstrate that fast-PACE can effectively extract good training data from daily interaction activities to build a reliable eye tracker with automatic updating and recalibration. The performance achieved is comparable to that from similar state-of-the-art methods. However, fast-PACE has the advantage that it automatically updates and recalibrates itself and is therefore able to adjust to variations in conditions over multiple days and sessions. Furthermore,

fast-PACE also shows considerable improvement over PACE [13] under the limitation of a small target dataset. This highly enhances the feasibility of using interactional cues for gaze learning in practical applications.

In future, we plan to investigate the impact of focus and attention on human eye gaze patterns and the performance of the gaze tracking system. Since it is not difficult to see how the underlying data validation method can be combined with other gaze estimation techniques, we also plan to investigate fast-PACE with appearance-based techniques using webcam video or for model-based techniques that work on infrared devices. In addition, we plan to extract and aggregate features from different perspectives to further improve the data validation mechanism as well as the gaze estimation [43, 51].

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